Cat-Eye Pumpkin — Assembly Guide

Oct 23, 2026 · @Ola

Before you start

You build the pumpkin as four bench sub-assemblies: base, mechanism, back shell, front mask. They are joined only at the end, after every motor and switch has been tested. Plan about 8–10 hours of assembly once everything is printed.

Build order

1. Prepare printed parts: inserts, magnets, reaming, tubes.
2. Stage 1: base, battery, power, neck spring.
3. Stage 2: mechanism frame, MG995, ratchet gear train.
4. Stage 3: plunger, ejector spring, black plate.
5. Stage 4: back shell — mechanism mounting, electronics, MG90S deadbolt, lid switch.
6. Stage 5: flame and LED band, fitted from above through the stem opening.
7. Stage 6: front mask and LCD eyes.
8. Stage 7: trapdoor lid.
9. Wiring, then first power-up and calibration with the helper sketch.
10. Final assembly and the first full scare cycle.

Directions used in this guide. Left, right, front and back are as seen standing in front of the pumpkin's face.

- The flame shaft sits on the head's centre line.
- The gear is left of the shaft; the rack teeth face left toward it.
- The MG995 sits behind the green frame.
- The lid hinges on the left and latches on the right.
- The ESP32 shelf is on the right; the IMU sits low on the left.

Safety

- The armed ejector spring throws the flame up at about 3 m/s. Never put your face or fingers over the stem opening while the lid is latched.
- Keep fingers out of the gear and rack whenever the 6 V rail is on. The MG995 stalls at about 2.5 A and will not stop for a finger.
- Insert the 18650 cells only after the BMS and the 5 A fuse are wired. Charge only with a 12.6 V 3S Li-ion charger.

Updated since the STL pack. While writing this guide I re-checked every screw and fastener, and found four fixes:

- **Stop screw:** now a flush M3×6 grub screw, with its nut trapped under the black plate. In the old plate the screw head stood above the plate, inside the flame's path at full over-travel.
- **Black plate screws:** counterbores are now 3.4 mm deep, so socket heads sit below the top face, under the flame.
- **Frame screw clearance:** the frame's upper-right mounting screw was hidden behind the black plate. It moved 18 mm lower, so all four frame screws can be reached from the front, and the back shell posts moved with it.
- **Back-wall screw:** moved 2.5 mm away from the rack channel so it can't bulge the wall into the carrier.

Print **black_plate.stl, backbone_frame.stl and head_back.stl from the updated v2C** pack linked in chat. All other STLs are unchanged.

The spring guide tube is **98.6 mm** long, not the 99.6 mm in the pack README.

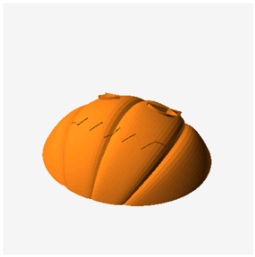
Parts & hardware checklist

Lay everything out before Stage 1. The build stalls most often on a missing M2 screw or magnet, not on a big part.

Printed parts (20 files, 21 pieces)

Part	Qty	Material
head_back, head_front	1 each	PLA, orange
trapdoor_lid	1	PLA, brown or orange
stem	1	PLA, green
lcd_retainer	2	PLA
base_shell	1	PLA or PETG, black
neck_mount	1	PETG
backbone_frame, channel_cover, black_plate	1 each	PETG
sector_gear, drive_disk, antireverse_pawl	1 each	PETG, 100 % infill
rack_carrier, shoulder_cap, keeper, deadbolt	1 each	PETG, 100 % infill
flame_body, flame_collar	1 each	Translucent PETG

Part	Qty	Material
tpu_pads (3 pads)	1 set	TPU 95A

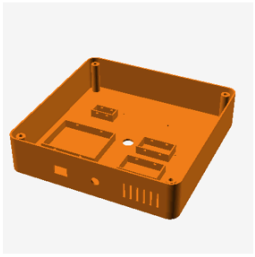

head_front
PLA

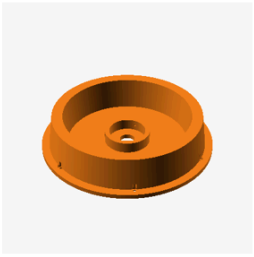

head_back
PLA

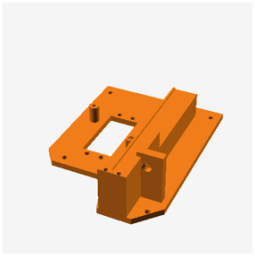

trapdoor_lid
PLA

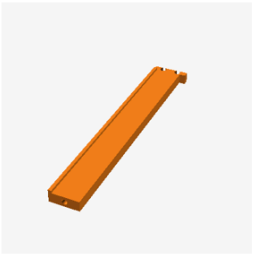

stem
PLA

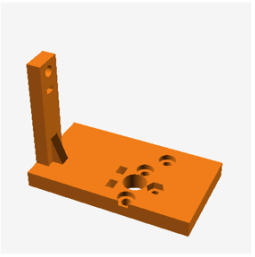

lcd_retainer
PLA x2

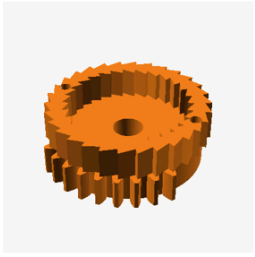

base_shell
PLA/PETG

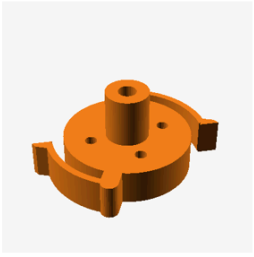

neck_mount
PETG

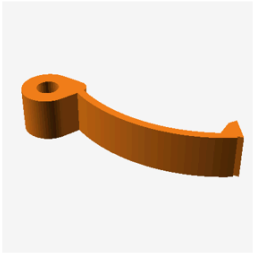

backbone_frame
PETG

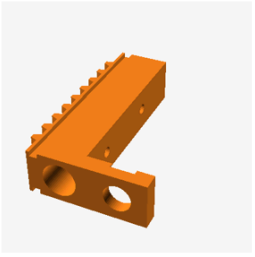

channel_cover
PETG

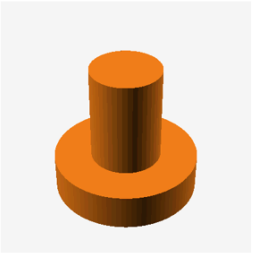

black_plate
PETG

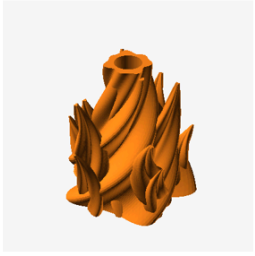

sector_gear
PETG

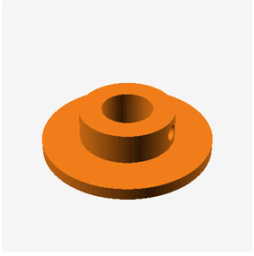

drive_disk
PETG

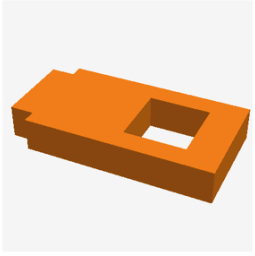

antireverse_pawl
PETG

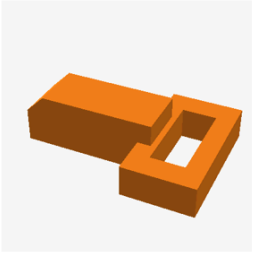

rack_carrier
PETG

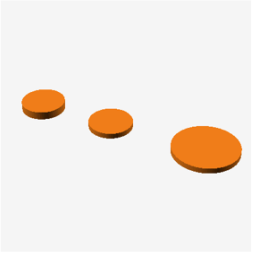

shoulder_cap
PETG


flame_body
clear PETG


flame_collar
clear PETG


keeper
PETG


deadbolt
PETG


tpu_pads
TPU 95A

Metal and springs

Item	Spec	Qty	Used in
Aluminium tube	Ø8 × 1 mm, cut to 165.6 mm	1	Flame shaft
Aluminium tube	Ø6 × 1 mm, cut to 98.6 mm	1	Ejector spring guide
Ejector spring	0.8 wire × 11.8 OD × 100 mm free	1	Mechanism
Neck spring	3.0 wire × 33 OD × 50 mm free	1	Neck

Item	Spec	Qty	Used in
Steel plate	230 × 230 × 4 mm, 4 holes Ø4.5 at ±106 mm from centre	1	Base ballast
Rubber feet	Ø20 stick-on	4	Under the steel plate

Fasteners, magnets, dowels

Item	Qty	Used in
M4 heat-set insert (Ø5.6 hole) + M4×10 screw	4 + 4	Steel plate to base
M3 heat-set insert (Ø4.1 hole, ~5 mm long)	4, +8 optional	Back-shell mechanism posts; expansion bays
M3×10 socket head	7	Frame to shell (4), black plate (3)
M3×12 + nut	4 + 4	MG995 to frame
M3×10 + nut	6 + 6	Neck mount to base
M3×10	3	Channel cover (2); use M3×12 for the anti-reverse pawl (1)
M3×6 grub screw	8	Stop (1), flame collar (1), rack carrier (2), neck cups (4)
M3 nut	1	Stop grub, trapped under the black plate
M3×25 + M3 large washer (DIN 9021)	1 + 1	Gear centre screw
M3×30	1	Lid hinge pin
M2×6 self-tapping	4	Servo horn to drive disk
M2×8	3	MG90S tabs (2), crank pin (1)
M2×12 self-tapping	2	Lid switch
M2.5×6 self-tapping	6	LCD retainers (4), IMU (2)
Neodymium magnet 8×3 mm	16	Shell seam (8 per half)
Neodymium magnet 6×2 mm	2	Lid hold-open
Neodymium magnet 3×2 mm	2	Gear index

Item	Qty	Used in
Steel dowel Ø3 × 10 mm (or cut nails)	4	Shell seam

Electronics

Item	Qty	Notes
ESP32-S3-DevKitC-1 N16R8	1	Headers must point up when mounted
GC9A01 1.28″ round LCD	2	Round PCB ⩽ Ø38.9 with header tab
MG995 servo, 180°	1	Keep the round disc horn
MG90S servo, 180°	1	Keep the single-arm horn
MPU6050 (GY-521)	1	
A3144 hall sensor	1	Needs 5 V supply
KW11-3Z lever micro-switch	1	Lid-closed sensor
WS2812B-2020 LEDs	4	4 LEDs cut from a 2020 strip, about 25 mm
74AHCT125 (DIP-14)	1	3.3 V to 5 V LED data; one gate used
Active buzzer, 5 V	1	
BC337 NPN transistor + 1 kΩ resistor	1 + 1	Buzzer driver
18650 cells, same model	3	
3S 18650 holder + 3S BMS (⩾ 10 A)	1 + 1	
XL4015 5 A buck module	1	Set to 6.0 V, servo rail
LM2596 buck module	1	Set to 5.0 V, logic rail
Blade fuse holder + 5 A fuse; inline 3 A fuse	1 + 1	Main; expansion line
Rocker switch, 19 × 13 mm panel cut-out	1	
DC panel jack 5.5×2.1 (11 mm hole) + 12.6 V 3S charger	1 + 1	Charging
1000 µF 16 V capacitor	1	Across the servo supply
1N5817 Schottky diode	1	ESP32 5 V feed

Item	Qty	Notes
Resistors 100 kΩ, 22 kΩ, 330 Ω	1 each	Battery sense, LED data
Silicone wire: 18 AWG (5 colours), 26 AWG, 30 AWG	—	Power and umbilical; signals; flame LEDs
Dupont F-F jumpers, heat-shrink, zip ties	—	

The second micro-switch from the Phase 1 BOM (deadbolt home) has no mount in the CAD and is not used. GPIO 42 stays free.

Tools & consumables

A multimeter and calipers are the two tools you cannot skip. One sets the buck voltages; the other sets the stop height.

- Soldering iron, solder, flux; a flat or insert tip for heat-set inserts (about 220 °C for PLA, 245 °C for PETG)
- Multimeter
- Digital calipers with a depth rod
- Hex keys 1.5 mm (M3 grubs) and 2.5 mm (M3 socket heads); small Phillips screwdriver
- Drill bits 1.6, 2.0, 3.2 and 8.0 mm, used by hand or in a slow drill
- Fine hacksaw or pipe cutter, a file, and a deburring tool for the aluminium tubes
- Needle-nose pliers, tweezers, flush cutters, wire strippers
- Thin and gel CA glue, 5-minute epoxy, a hot-glue gun
- Kapton tape, thin double-sided foam tape, zip ties (2.5 mm wide)
- A spring-type clothes peg or a small binder clip, used to hold the plunger down in Stage 5
- A permanent marker for magnet polarity
- A computer with Arduino IDE 2.x, the Espressif ESP32 core 3.x, and the libraries ESP32Servo and Adafruit NeoPixel

Prepare the printed parts

Do all inserts, magnets and fit checks now, while the parts are loose. Most later problems come from a bore that is 0.1 mm too tight.

Clean up and fit-check sliding parts

1. Remove the tree supports from both shell halves and the lid knuckles. Inside surfaces can stay rough.
2. Run an 8.0 mm drill through the rack carrier bore by hand. The Ø8 aluminium tube should push in firmly without a hammer.
3. Slide the Ø8 tube through the flame collar, the flame body and the black plate bushing. All three must slide freely without wobble. Clean them with the 8.0 mm drill if not.
4. Put the sector gear on the drive disk's hub. It must spin freely with no rub. If it drags, sand the hub lightly with 400-grit.
5. Slide the Ø6 tube through the carrier's lug hole: free. Push it into the frame's shelf hole and the black plate's blind socket: firm push-fit.

Cut the tubes

1. Cut the Ø8×1 tube to **165.6 mm** (flame shaft).
2. Cut the Ø6×1 tube to **98.6 mm** (spring guide).
3. Square and deburr both ends inside and out. A burr on the shaft scores the black plate bushing.

Heat-set inserts

1. Base: 4 × M4 inserts into the corner bosses, from the open underside.
2. Back shell: 4 × M3 inserts into the front ends of the four mechanism posts. Keep them square; the frame bolts flat against these faces.
3. Optional: 8 × M3 into the two expansion-bay platforms (smoke bay right, motor bay left).

Shell seam magnets and dowels

The two halves only snap together if every magnet pair attracts.

1. Stack all 16 magnets (8×3 mm) into one column. Mark the exposed top face with the marker.
2. Back shell: epoxy 8 magnets into the seam pockets, **marked face out**.
3. Front mask: epoxy 8 magnets into its pockets, **marked face in** (unmarked face out).
4. Glue the 4 Ø 3 × 10 mm dowels into the back shell's dowel holes, 5 mm in and 5 mm proud.
5. Before the epoxy sets, put cling film over the back-shell seam and press the mask on. This sets the magnets flush. Separate the halves after 10 minutes.

Small magnets and pads

1. Lid hold-open: one 6×2 mm magnet in the pocket on the lid's top surface near the hinge, one in the matching pocket on the crown's stop block. Check they attract when the lid is folded open, then glue.

2. Gear index: keep the two 3×2 mm magnets loose for now. Their polarity is set in Stage 2.
3. TPU pads: press the Ø12 × 1 pad into the lid's underside pocket and the Ø8 × 1 pad into the shoulder cap's pocket. Keep the Ø8 × 1.5 stop pad for Stage 3.

Steel plate

Have the plate drilled with 4 × Ø4.5 holes at ±106 mm from centre in both directions. Or clamp the base on it and mark through the boss holes.

Stage 1 — Base, power and neck spring

The base carries the battery, both buck converters and 1.8 kg of steel, and sends five power wires up the neck. Finish it and prove both voltages before anything else is connected.



Base upside down on the bench. Largest cradle: 18650 holder. Two medium cradles: XL4015 and LM2596. Narrow cradle: fuse holder. Rocker, DC jack and vents are in the back wall.

Power wiring

1. Turn the base upside down. Strap each part into its cradle with a zip tie through the cradle's side slots:
   - 3S 18650 holder in the largest cradle
   - XL4015 module in the larger of the two medium cradles
   - LM2596 module in the smaller medium cradle
   - blade-fuse holder in the narrow cradle
2. Press the rocker switch into the rectangular cut-out and the DC jack into the round hole in the back wall.
3. Wire the cell holder to the BMS (B−, B1, B2, B+) exactly as printed on the BMS. Stick the BMS to the top skin beside the holder with foam tape.
4. Main line in 18 AWG: BMS P+ → 5 A fuse → rocker switch → a junction. From the junction:
   - XL4015 IN+
   - LM2596 IN+
   - the 3 A inline fuse → the VBAT_EXP umbilical wire
5. BMS P− goes to both modules' IN−.
6. Charging: DC jack centre pin → BMS P+, sleeve → BMS P−, ahead of the rocker so you can charge with the power off. Label the jack "12.6 V Li-ion only".

Set the voltages

1. Check that the three cells are within 0.05 V of each other, then insert them.
2. Switch on with nothing connected to the outputs.
3. Turn the LM2596 pot until its output reads **5.00 V**.
4. Turn the XL4015 voltage pot to **6.00 V**. Turn its current pot fully clockwise (maximum, about 5 A). The MG995 stalls at about 2.5 A, and a lower limit makes it twitch.
5. Switch off.

Umbilical

1. Cut five 18 AWG silicone wires, 350 mm each:
   - red: +5 V, from the LM2596 OUT+
   - orange: +6 V, from the XL4015 OUT+
   - yellow: VBAT_EXP, from the 3 A fuse
   - two black: GND, one from each module's OUT−

2. Solder them on, twist them loosely, and thread them up through the Ø16 centre hole in the base top.

Neck mount and spring

1. Thread the umbilical through the neck mount's Ø14 centre hole and set the mount on the base, holes aligned.
2. From inside the base, drive 6 × M3×10 up through the top skin and the flange into M3 nuts in the hex traps on the flange.
3. Slide the neck spring over the umbilical and seat it 10 mm deep in the cup. Snug 2 × M3×6 grubs against the bottom coil. They only stop it turning; don't crush the wire.
4. After the power test, close the base: 4 rubber feet under the steel plate, then 4 × M4×10 into the corner bosses.

Stage 2 — Mechanism frame, MG995 and the ratchet gear train

The gear only ever turns **clockwise seen from the front**. The MG995 drives it through two flexure pawls and resets backwards through the ratchet while the gear stays parked. So the servo's direction and zero are found first, before the horn goes on.

Find the servo direction and zero (bare spline)

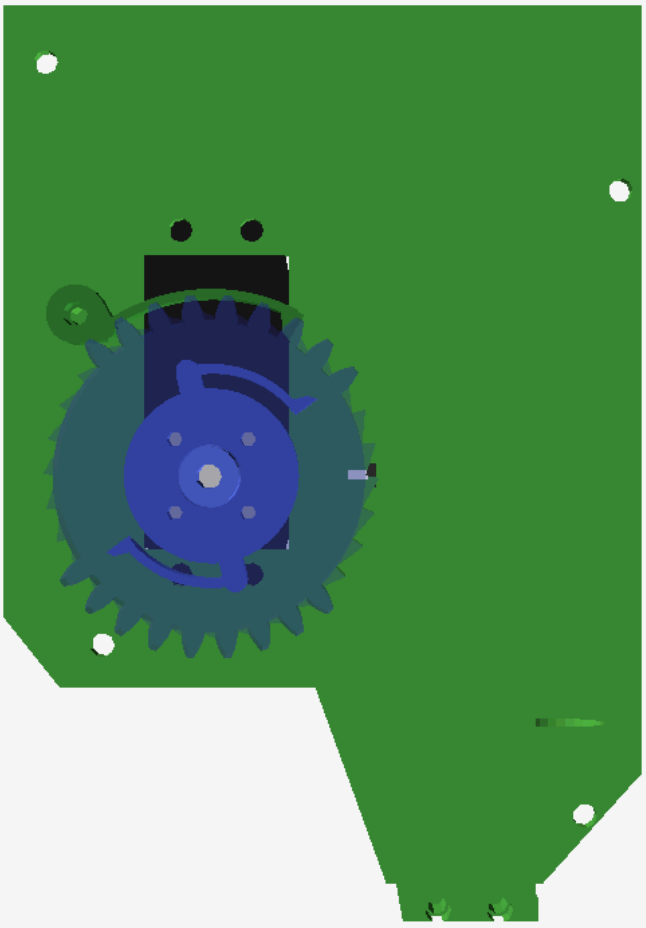
1. Bench-wire the MG995: brown → GND, red → +6 V (the base's XL4015, or a 6 V / 3 A bench supply), orange → GPIO 17. Power the ESP32 from USB and join its GND to the 6 V supply's GND.
2. Flash the helper sketch from *First power-up & calibration*.
3. Send `w`. The pulse steps from 1100 to 1500 µs and the spline turns about 35°. Watch it end-on, as from the pumpkin's front.
   - Clockwise: leave it.
   - Anticlockwise: send `v` once. This sets `reversed=1`; write it down for the firmware.
4. Send `a0`. The spline is now at logical zero. Don't turn it by hand from here on.

Frame, servo and hall sensor

1. Push the MG995 through the frame window from the back, spline toward the front, tabs flat on the frame's rear face. Bolt it with 4 × M3×12, nuts on the front face. Leave out the rubber grommets; they let the servo rock under load.
2. Push the A3144 into the small pocket on the post right of the servo spline (3 o'clock), branded face toward the front. Feed the legs through the slot to the back.
3. Solder three 26 AWG wires, about 300 mm, on the back. With the branded face toward you the legs are, left to right: VCC (+5 V), GND, OUT (GPIO 47). Sleeve each joint.

Horn and drive disk

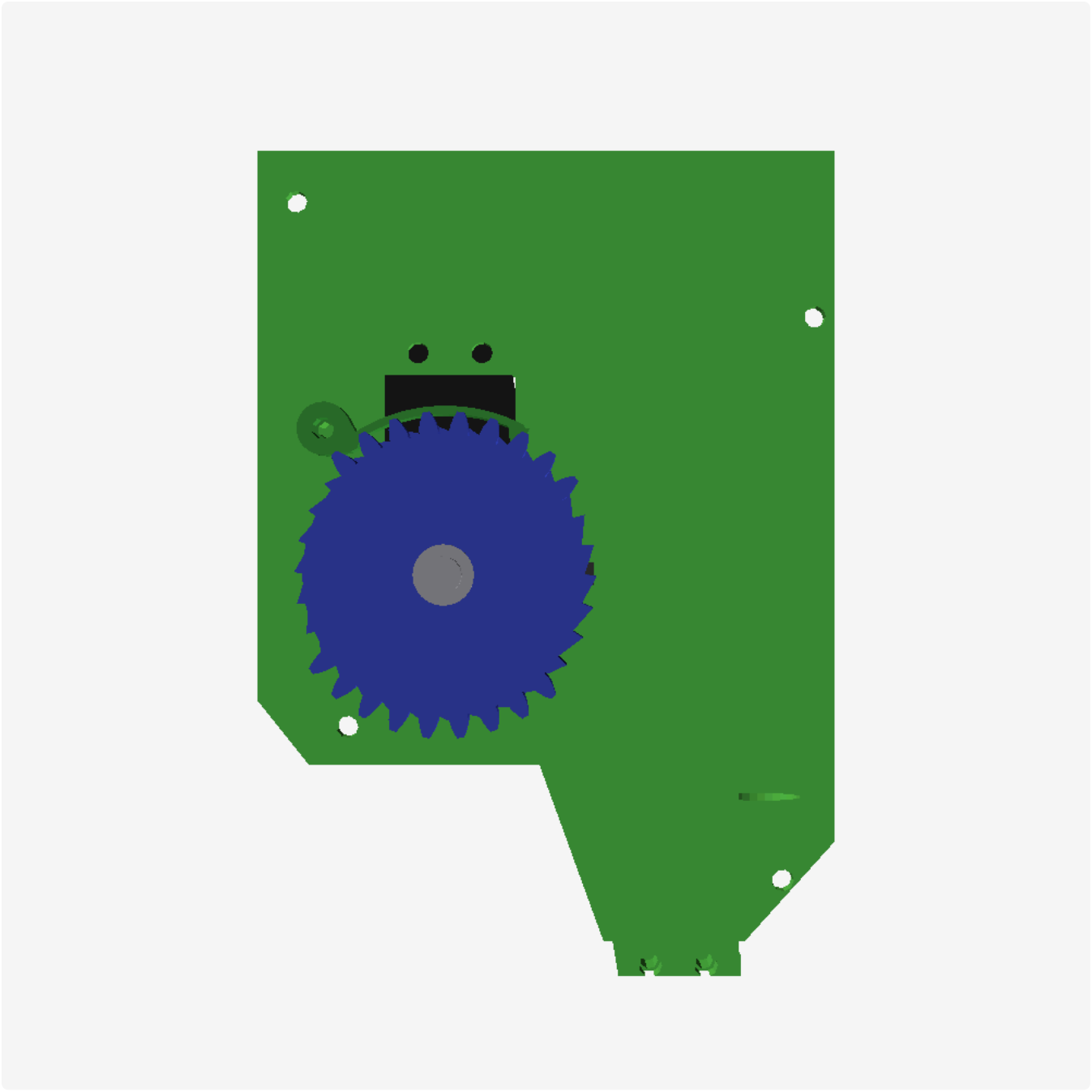
1. Lay the MG995 round horn on the drive disk's flat back face (the side without the hub). Push the M3×25 through the hub and the horn centre to align them.
2. If two or more horn holes line up with the disk's four Ø2.2 holes, use those. If not, drill Ø1.6 mm through two opposite horn holes, 5 mm into the disk.
3. Drive M2×6 self-tappers **from the horn's spline side** into the disk. No screw head may stand on the disk's front, where the gear sits.
4. With the servo still at `a0`, press the horn onto the spline. The two pawl lugs should point to about 2 o'clock and 8 o'clock, seen from the front. Use the spline tooth closest to that.



Front view, gear drawn transparent. The drive disk's two curved pawls sit inside the gear's ratchet cup; the anti-reverse pawl hangs from its post at 11 o'clock.

Index magnets, gear and anti-reverse pawl

1. Find the magnet face the hall sensor sees: wire the sensor (VCC → +5 V, GND, OUT → GPIO 47), hold a 3×2 mm magnet on its branded face and send `s`. The face that shows `hall=MAGNET` is the one to mark.
2. Glue both magnets into the two pockets in the gear's ratchet-cup rim, marked face out, with thin CA.
3. Slide the gear onto the hub, ratchet cup toward the servo. Wiggle it until the drive pawls drop into the cup's inner teeth.
4. Fit the M3 large washer and drive the M3×25 into the servo shaft, firmly. If it bottoms out before it clamps, cut 2 mm off.
5. Put the anti-reverse pawl's eye on the round post at 11 o'clock, its lug resting in the gear's outer teeth at about 1 o'clock. Fix it with M3×12 into the post. The lug should press lightly into the teeth.
6. Turn the gear clockwise by hand until a magnet faces the sensor and `s` shows `hall=MAGNET`. This is the **parked** position: the gear's toothless arc faces right, where the rack will run.



Parked: no gear teeth between 1 and 5 o'clock on the rack side.

Check the ratchet and set the zero trim

1. By hand, the gear must click freely clockwise and lock anticlockwise. If it locks both ways, a pawl is too tight: sand the pawl lug's sloped face.
2. With the gear parked, send `a1`, `a2`, `a3`, … and watch the gear. Note the first angle where it moves.
3. Send `t` with that angle minus 1, for example `t3`. Write the trim down for the firmware.
4. If the gear moves at `a1` already, refit the horn one spline tooth anticlockwise and repeat.

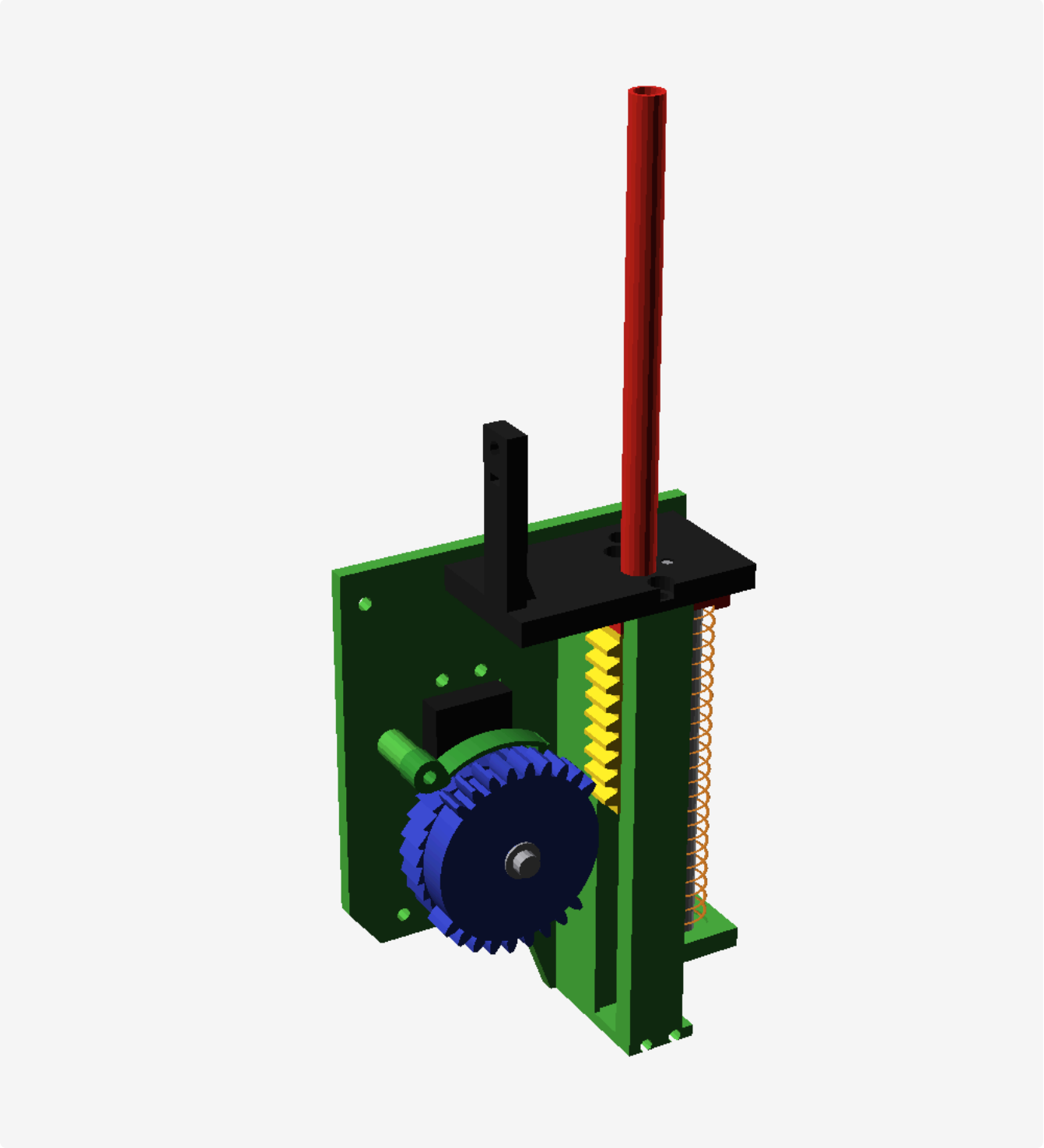
5. Push the gear back anticlockwise until it stops on the anti-reverse pawl; it is parked again.

Channel cover

Hold the cover on the front of the channel, guide ridge facing in, its foot under the channel's bottom block. Fix it with 2 × M3×10 into the bottom block.

Stage 3 — Plunger, ejector spring, black plate and stop screw

The plunger is the carrier, rack and Ø8 tube, pushed up by the ejector spring against an adjustable stop grub in the black plate. The grub sets how high the rack's first tooth sits, which decides whether the gear picks it up cleanly. Keep the gear **parked** for this whole stage.



After Stage 3: plunger at the top stop, ejector spring on its guide tube to the right of the channel, black plate with the wire-anchor tab on the left.

1. **Tube into carrier.** Push the Ø8 tube into the carrier bore until its bottom end is flush with the carrier's bottom face. Tighten 2 × M3×6 grubs through the holes in the carrier's right face, opposite the teeth. Optional: spot-drill 0.5 mm dimples in the tube under the grubs.
2. **Stop nut and grub.** Press an M3 nut into the hex pocket on the black plate's **underside** and tack it with a drop of CA on its edge. Screw the M3×6 grub in from the top with the 1.5 mm key until its tip stands **0.5 mm below** the plate's underside. Measure with the

calipers' depth rod. This is the nominal stop.

3. **Stop pad.** Slide the plate onto the tube and rest it on the carrier's lug, the wide flat block on top of the carrier. Mark the lug through the stop-grub hole, then lift the plate off. Glue the Ø8 × 1.5 TPU pad centred on the mark.
4. **Spring and guide.** Push the Ø6 guide tube down through the shelf hole (the green shelf right of the channel) until its bottom end is flush with the shelf's underside. Drop the ejector spring over it, seated in the shelf's round recess.
5. **Carrier into the channel.** Lower the carrier into the channel from the top: rear groove onto the frame's ridge, front groove onto the cover's ridge, rack teeth facing the gear, lug hole over the guide tube. Push it down to compress the spring and hold it.
6. **Black plate.** Keep the carrier pushed down about 10 mm. Slide the plate down the tube, anchor tab on the left, until the guide tube's top enters the plate's blind socket underneath and the plate sits flat on the wall tops. Fix it with 3 × M3×10 socket heads in the counterbores. Let the carrier rise; it stops with the pad against the grub.

Checks before moving on

- Push the tube down to the bottom and let go a few times. It returns smoothly, and the rack never touches the parked gear.
- Lay a straight edge across the black plate. It must touch only plastic: the three screw heads and the grub sit below the top face. At full over-travel the flame passes **1.6 mm** above this face.
- Leave the pick-up check (gear catching the first rack tooth) for calibration. It needs the servo powered.

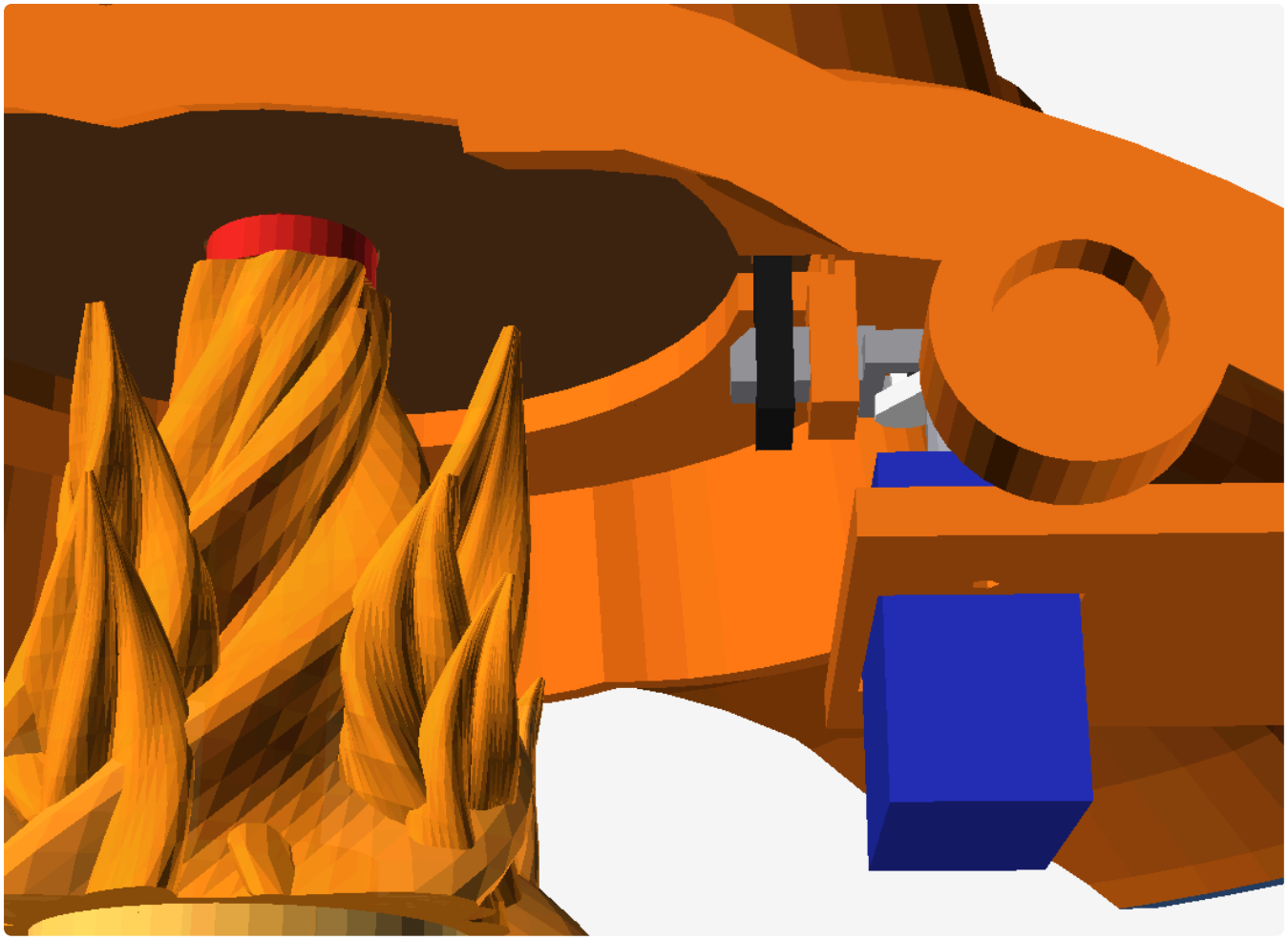
Stage 4 — Back shell: mechanism, electronics, deadbolt, lid switch

The mechanism slides into the back shell from the open front and bolts to four posts. Fit the small parts inside the shell first, while there is room. Lay the shell on its back on a towel, seam facing up.

IMU and ESP32

1. **IMU.** Fix the GY-521 on the low rectangular pad left of the neck cup, chip side up, flat. Use 2 × M2.5×6 if its holes line up with the pilots, otherwise thin foam tape. It must not wobble. Solder five 26 AWG wires, about 250 mm: VCC, GND, SCL, SDA, INT.
2. **ESP32.** Put the DevKitC on the shelf on the right, resting on its two rails, **header pins pointing up** and USB-C ports toward the head's centre. If your headers are soldered underneath, mount the board upside down. Strap it with two zip ties or foam tape.

MG90S and deadbolt



Under the crown, right side: the keeper hangs from the closed lid, the grey deadbolt passes through it, and the MG90S (blue) sits on the shelf below with its crank driving the bolt's yoke.

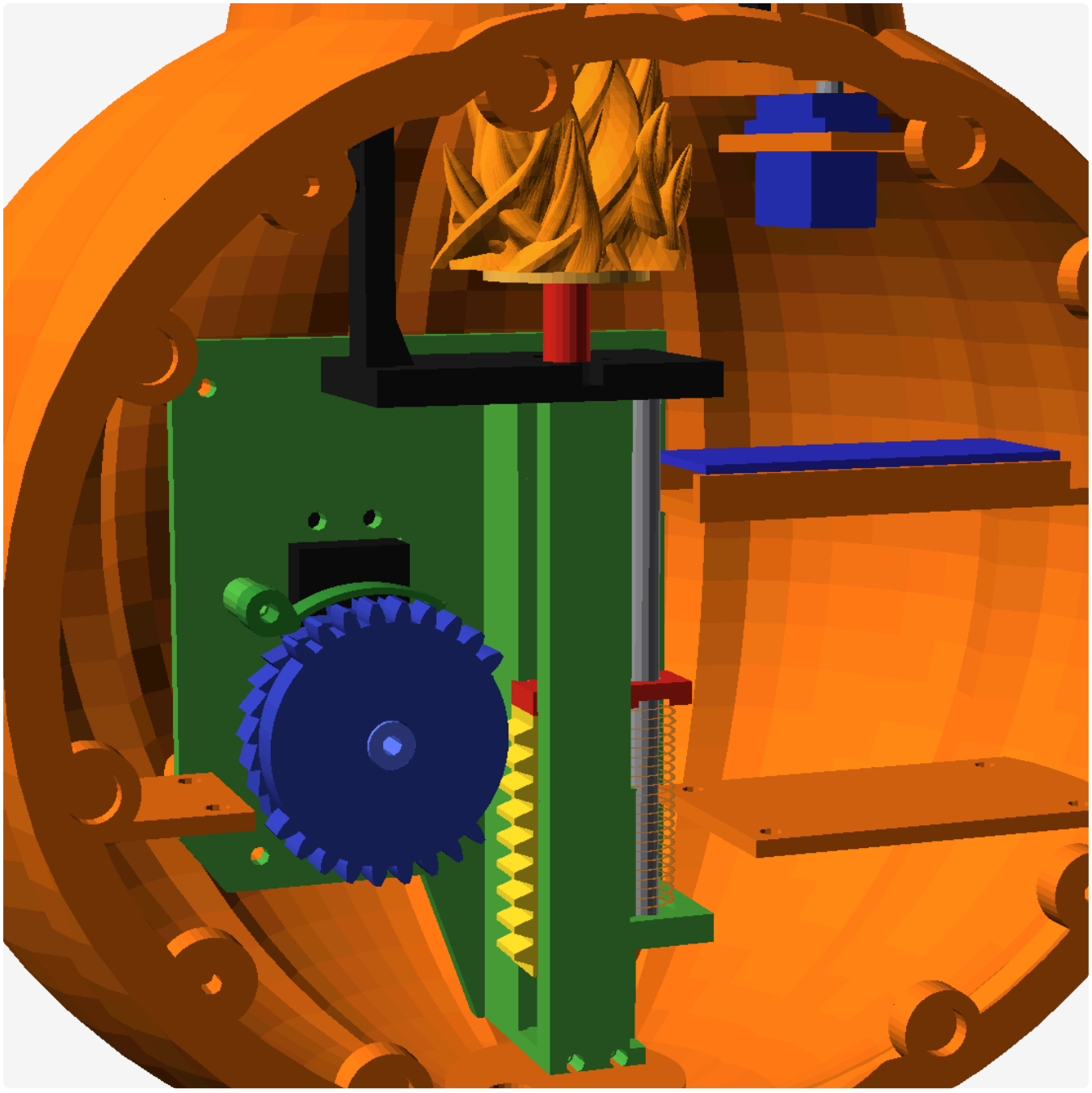
1. **Crank pin.** On the MG90S single-arm horn, drill a Ø1.6 mm hole exactly **5.0 mm** from the spline centre. Drive an M2×8 up through it from the spline side so about 5 mm of thread stands above the arm.
2. **Servo.** Drop the MG90S into the rectangular hole in the shelf under the crown on the right, spline up and toward the front. Fix the tabs with the two servo screws.
3. **Bolt.** From the right, slide the deadbolt's bar into its guide under the crown ledge. The square yoke frame stays outside the guide, its slot above the servo spline.
4. **Arm.** Wire the MG90S (brown GND, red +6 V, orange GPIO 18) and send `b90`. Fit the arm pointing **straight back**, pin in the yoke slot, and fix it with the horn screw.
5. **Find LOCK and OPEN.** Send `b45`, then `b135`. The one that slides the bolt fully **left**, toward the keeper, is LOCK. The other is OPEN. Store them with `L45` / `O135` (or swapped) and write both down for the firmware.
6. Leave the bolt at OPEN (`b<open>`) until the lid is fitted.

Lid switch

1. The KW11-3Z goes into the block under the ledge at the back left of the stem opening. Insert it from below, lever up through the slot in the ledge.
2. Fix it with 2 × M2×12 self-tappers through the block's side holes.
3. Wire COM → GND and NO → GPIO 41. The released lever must stand about 1–2 mm above the ledge so the closed lid presses it.

Mechanism into the shell

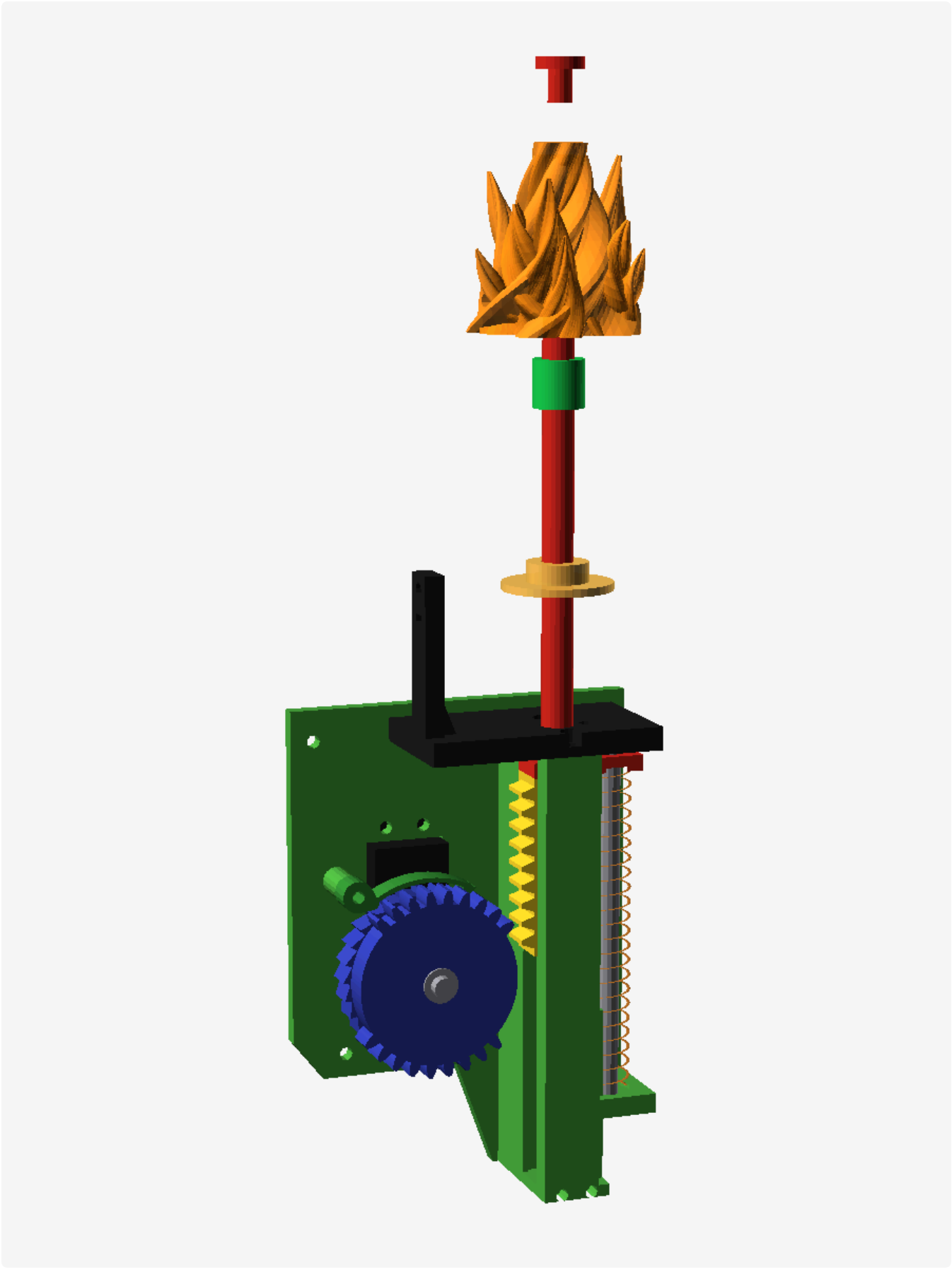
1. Push the flame tube fully down and clip a clothes peg on it just above the black plate. The tube's top is now below the crown ledge, so it can pass under it.
2. Gather the MG995 and hall-sensor wires behind the frame.
3. Slide the mechanism in from the front, servo first. The frame's rear face lands flat on the four posts, and the tube sits centred under the stem opening.
4. Drive 4 × M3×10 through the frame's mounting holes into the post inserts, with the 2.5 mm key from the front. All four are reachable with the black plate fitted.
5. Hold the tube and remove the peg. Let the tube rise slowly through the stem opening to its top stop.



Inside the back shell, plunger armed: gear and rack centre left, anchor tab and flame above the black plate, ESP32 shelf on the right, MG90S under the crown at top right.

Stage 5 — Flame, LED band and wire loop

The flame goes on last, from above through the stem opening, with the plunger at its top stop. In that position 44 mm of tube stands above the crown. The stack on the tube is, bottom to top: collar, LED band, flame body, shoulder cap.



Exploded flame stack on the tube: collar (bottom), LED band (green), flame body, shoulder cap (top).

1. **LED band (bench).** Cut 4 LEDs from the WS2812B-2020 strip, about 25 mm, which is one wrap of the tube. Solder three 30 AWG silicone wires, 200 mm, to the input end: +5 V, GND, DIN. Test it with `f` once it is wired (see *Wiring & power distribution*).
2. **Collar.** Slide the collar down over the tube top, flange down. Set its underside **47.5 mm below the tube's top end** with the calipers. Tighten its grub with an L-key through the stem opening.
3. **Band.** Wrap the band round the tube with its top edge **28–30 mm below the tube's top end**, LEDs facing out and wires leaving its lower edge. Hold it with Kapton tape.
4. **Body.** With tweezers, feed the three wires into the flame body's bottom opening and out through the small side hole. Slide the body down over the band, pulling the wires through as it goes, until it sits on the collar.
5. Turn the body so the side hole faces **left**, toward the anchor tab.
6. **Cap.** Push the shoulder cap's plug into the tube top, pad side up. The body is now clamped between collar and cap. If it rattles, reset the collar 0.5 mm higher. A dot of hot glue between the body and the collar's flange stops it turning.
7. **Wire loop.** With the plunger still at its top stop, thread the wires through the anchor tab's Ø4 hole from the flame side. Leave them just slack, about 40 mm between the flame's side hole and the tab. Zip-tie them through the slot below the hole and run them down the tab toward the ESP32.
8. **Loop test.** Push the flame fully down by hand and let it rise, five times. The loop must stay outside the flame and never fall under it onto the black plate. Adjust the length by 5 mm if it does.

Stage 6 — Front mask and LCD cat eyes

Each round LCD drops into a socket behind its cat-eye window and is held by a printed retainer ring. The almond window is smaller than the display, so only the slit-pupil area shows. In this guide, **L** is the eye on your left as you face the pumpkin.

1. Lay the mask face down on a towel. Handle the LCDs by their PCB edges; the glass corners chip easily.
2. Drop each module into its socket glass first, header tab pointing **down** into the socket's notch.
3. Lay a retainer ring over the PCB, its notch over the header tab. Fix it with 2 × M2.5×6 self-tappers into the socket ears, snug only.
4. Plug 7 female-female jumpers, 20–30 cm, onto each module's header: VCC, GND, SCL, SDA, DC, CS, RST. Put a dab of hot glue over the jumper housings so vibration can't shake them off.
5. Label the bundles **L** and **R**.

6. Run both bundles along the inside of the mask toward the seam on the right, next to the ESP32 shelf. Leave enough slack to lay the mask face down beside the head with everything connected, for servicing.

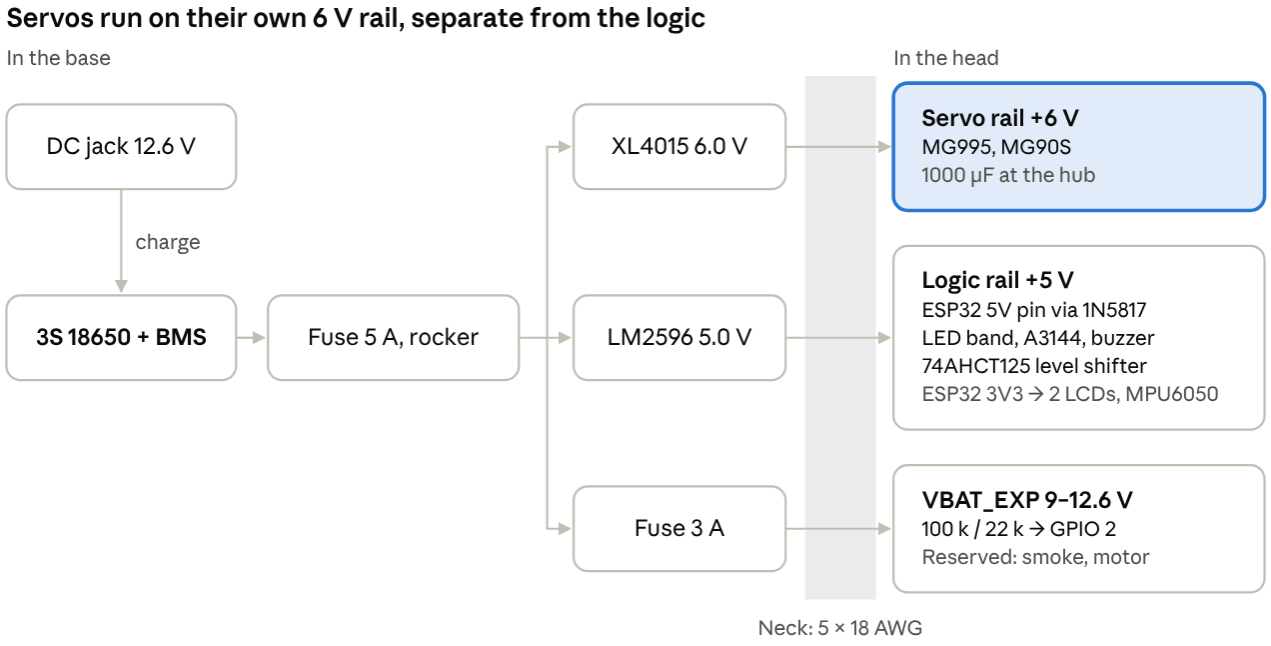
Stage 7 — Trapdoor lid, keeper, stem and hinge

The lid hinges on the left of the crown and is held shut by the deadbolt through a keeper on its right edge. With the flame fitted the plunger sits at its top stop and the lid cannot close yet. The closing and locking tests happen in calibration, after the servo pulls the flame down.

1. **Keeper.** Epoxy the keeper's tenon into the slot in the lid's underside, on the edge opposite the knuckles. Its rectangular window faces left-right, along the bolt's path.
2. **Stem.** Glue the stem's plug into the boss on top of the lid. The cone under the stem sits in the boss's countersink.
3. **Pads.** Check the Ø12 TPU pad is flush in its pocket on the lid's underside.
4. **Hinge.** Set the lid's two knuckles either side of the shell's knuckle at the left of the crown. Drive the M3×30 from the front through all three. It self-taps into the shell knuckle; the lid knuckles turn on its shank. Stop when the head is flush.
5. **Swing check.** The lid must swing freely through its full arc. Opened fully it stops at about 110° on the stop block and the two 6×2 mm magnets hold it there. If it binds, back the hinge screw out a quarter turn.

Wiring & power distribution

Everything in the head is wired to one small hub board: power buses, servo connectors, the LED level shifter and the buzzer driver. The ESP32 connects to the hub with jumpers. All grounds join at the hub, and the servos never feed from the ESP32's 5 V pin.



power distribution · base to head

Ground is not drawn: both black umbilical wires and every device ground join at the hub.

Build the hub on a 50 × 70 mm perfboard. Stick it with thick foam tape to the inside of the back shell, right side, behind the ESP32 shelf; that corner is empty.

- **GND bus:** both black umbilical wires, plus every ground below.
- **+6 V bus:** orange umbilical wire, two 3-pin servo headers (signal, +6 V, GND), and the 1000 µF capacitor across +6 V and GND (stripe to GND).
- **+5 V bus:** red umbilical wire, plus the 1N5817 to the ESP32: anode on the bus, banded end to the ESP32's **5V** pin. The diode lets USB and battery be connected together.
- **Level shifter:** 74AHCT125 (DIP-14) for the LED data.
  - pin 14 → +5 V; pin 7 → GND
  - pin 1 (enable) → GND
  - pin 2 ← GPIO 39; pin 3 → 330 Ω → LED DIN
  - unused enables, pins 4, 10, 13 → +5 V
- **Buzzer driver:** BC337 (flat face toward you: collector, base, emitter).
  - buzzer + → +5 V; buzzer − → collector
  - emitter → GND
  - base ← 1 kΩ ← GPIO 40
- **Battery sense:** yellow VBAT_EXP wire → 100 kΩ → GPIO 2, and 22 kΩ from GPIO 2 to GND. Insulate the rest of VBAT_EXP; it is reserved for the smoke machine and motor.

Pin map

Device	Device pin	Goes to
LCD L (your left)	VCC / GND	ESP32 3V3 / GND
LCD L	SCL / SDA / CS / DC	GPIO 12 / 11 / 10 / 13
LCD R	VCC / GND	ESP32 3V3 / GND
LCD R	SCL / SDA / CS / DC	GPIO 7 / 6 / 5 / 4
LCD L and R	RST (shared)	GPIO 14
MG995	orange / red / brown	GPIO 17 / +6 V / GND
MG90S	orange / red / brown	GPIO 18 / +6 V / GND
MPU6050	VCC / GND	ESP32 3V3 / GND
MPU6050	SCL / SDA / INT	GPIO 9 / 8 / 16
A3144 hall	VCC / GND / OUT	+5 V / GND / GPIO 47
Lid switch	COM / NO	GND / GPIO 41
LED band	+5 V / GND / DIN	+5 V / GND / 330 Ω from 74AHCT125 pin 3
74AHCT125	pin 2	GPIO 39
Buzzer driver	1 kΩ to BC337 base	GPIO 40
Battery sense	divider midpoint	GPIO 2
Reserved	smoke / motor IN1, IN2 / spare	GPIO 15 / 1, 48 / 42, 21

Rules that prevent most faults

- Use 18 AWG for every +6 V and GND run to the servos, and keep them short.
- Never power a servo from the ESP32. An MG995 stall browns it out instantly.
- The hall sensor needs 5 V; its output only pulls low, so it is safe on a 3.3 V pin.
- Before the first battery power-up, check with the multimeter that +6 V, +5 V and 3V3 each read at least a few hundred ohms to GND.

First power-up & calibration

Run the helper sketch below before the show firmware. It tests every motor, switch and LED, finds the four numbers the firmware needs (`REVERSED`, `TRIM`, `LOCK_DEG`, `OPEN_DEG`), and runs one full scare cycle.

Arduino IDE setup

1. Boards Manager: install **esp32 by Espressif** 3.x. Library Manager: install **ESP32Servo** and **Adafruit NeoPixel**.
2. Tools: board *ESP32S3 Dev Module*, *USB CDC On Boot: Enabled*, *Flash Size: 16MB*, *PSRAM: OPI PSRAM*.
3. Upload the sketch and open the Serial Monitor at 115200 baud with line ending *Newline*.

```
// Cat-Eye Pumpkin - assembly & calibration helper (not the show firmware)
// Board: ESP32S3 Dev Module, USB CDC On Boot: Enabled. Libraries:
ESP32Servo, Adafruit NeoPixel.
// Serial Monitor: 115200 baud, line ending "Newline".
#include <ESP32Servo.h>
#include <Adafruit_NeoPixel.h>

// ---- write your calibration results here so they survive a reboot ----
bool  REVERSED = false;   // MG995 direction  (command v)
float TRIM     = 0;       // MG995 zero trim, degrees (command t)
int   LOCK_DEG = 45;      // MG90S deadbolt locked (command L)
int   OPEN_DEG = 135;     // MG90S deadbolt open   (command O)

const int PIN_MG995 = 17, PIN_MG90S = 18, PIN_LID = 41, PIN_HALL = 47;
const int PIN_LED = 39, PIN_BUZZ = 40, PIN_VBAT = 2;

Servo cock, latch;
Adafruit_NeoPixel flame(4, PIN_LED, NEO_GRB + NEO_KHZ800);
float cockPos = 0;        // last logical MG995 angle

void cockWrite(float logical) {
  float s = logical + TRIM;
  if (REVERSED) s = 180 - s;
  s = constrain(s, 0.0f, 180.0f);
  if (!cock.attached()) cock.attach(PIN_MG995, 500, 2500);
  cock.writeMicroseconds(500 + (int)(s * 2000.0f / 180.0f));
  cockPos = logical;
}

void cockMove(float target, int msPerDeg) {      // slow, 1 degree per step
  while (fabsf(target - cockPos) >= 1.0f) {
```

```
    cockWrite(cockPos + (target > cockPos ? 1.0f : -1.0f));
    delay(msPerDeg);
  }
  cockWrite(target);
}

void latchWrite(int deg) {
  if (!latch.attached()) latch.attach(PIN_MG90S, 500, 2500);
  latch.write(constrain(deg, 0, 180));
}

bool lidClosed()  { return digitalRead(PIN_LID) == LOW; }
bool magnetSeen() { return digitalRead(PIN_HALL) == LOW; }
float vbat()      { return analogReadMilliVolts(PIN_VBAT) / 1000.0f * (122.0f
/ 22.0f); }

void flameOn(bool on) {
  for (int i = 0; i < 4; i++) flame.setPixelColor(i, on ? flame.Color(255,
70, 0) : 0);
  flame.show();
}

void beep(int ms) { digitalWrite(PIN_BUZZ, HIGH); delay(ms);
digitalWrite(PIN_BUZZ, LOW); }

void status() {
  Serial.printf("lid=%s hall=%s vbat=%.2fV | cock=%.0f reversed=%d trim=%.1f
lock=%d open=%d\n",
                lidClosed() ? "CLOSED" : "open", magnetSeen() ? "MAGNET" : "-
", vbat(),
                cockPos, REVERSED ? 1 : 0, TRIM, LOCK_DEG, OPEN_DEG);
}

void waitLidClosed() {                          // closed and steady for 250
ms
  uint32_t since = 0;
  while (true) {
    if (lidClosed()) { if (!since) since = millis(); if (millis() - since >
250) return; }
    else since = 0;
    delay(10);
  }
}

void fullCycle() {                              // the 5-step hand-off,
start from ARMED
```

```
  Serial.println("FIRE: bolt open");
  flameOn(true); beep(300); latchWrite(OPEN_DEG);
  delay(3000); flameOn(false);
  Serial.println("Cock 0 -> 134 (2 deg past the 132 notch), hold on the
pawl");
  cockMove(0, 4); cockMove(134, 8); delay(300); cock.detach();
  Serial.println("Close the lid...");
  waitLidClosed();
  latchWrite(LOCK_DEG); delay(300);
  Serial.println("Release 132 -> 180, then reset 180 -> 0");
  cockMove(180, 6); delay(200);
  cockMove(0, 3); delay(300);
  cock.detach(); latch.detach();
  Serial.println("ARMED");
}

void setup() {
  Serial.begin(115200);
  pinMode(PIN_LID, INPUT_PULLUP);
  pinMode(PIN_HALL, INPUT_PULLUP);
  pinMode(PIN_BUZZ, OUTPUT); digitalWrite(PIN_BUZZ, LOW);
  ESP32PWM::allocateTimer(0); ESP32PWM::allocateTimer(1);
  cock.setPeriodHertz(50); latch.setPeriodHertz(50);
  flame.begin(); flame.show();
  delay(1500);
  Serial.println("Pumpkin helper: a<deg> b<deg> t<trim> v w x f z s L<deg>
O<deg> F");
}

void loop() {
  if (!Serial.available()) return;
  String in = Serial.readStringUntil('\n'); in.trim();
  if (in.length() == 0) return;
  char c = in.charAt(0);
  float v = in.substring(1).toFloat();
  switch (c) {
    case 'a': cockMove(constrain(v, 0.0f, 180.0f), 8); break;      // MG995
to logical angle
    case 'b': latchWrite((int)v); break;                           // MG90S
to raw angle
    case 't': TRIM = v; cockWrite(cockPos); break;                 // zero
trim
    case 'v': REVERSED = !REVERSED; break;                         // flip
MG995 direction
    case 'w':                                                      //
direction test, bare spline only
```

```
      if (!cock.attached()) cock.attach(PIN_MG995, 500, 2500);
      for (int us = 1100; us <= 1500; us += 5) { cock.writeMicroseconds(us);
delay(10); }
      Serial.println("Seen from the front: clockwise -> keep; anticlockwise -
> send v");
      break;
    case 'x': cock.detach(); latch.detach(); break;                // both
servos limp
    case 'f': flameOn(true); delay(3000); flameOn(false); break;   // LED
test
    case 'z': beep(200); break;                                    // buzzer
test
    case 'L': LOCK_DEG = (int)v; latchWrite(LOCK_DEG); break;
    case 'O': OPEN_DEG = (int)v; latchWrite(OPEN_DEG); break;
    case 'F': fullCycle(); break;
    case 's': break;
    default: Serial.println("?"); break;
  }
  status();
}
```

Command	Does
`a<deg>`	MG995 to a logical angle, slowly (trim and direction applied)
`b<deg>`	MG90S to a raw angle
`t<deg>`	Set the MG995 zero trim
`v`	Flip the MG995 direction
`w`	Direction test, bare spline only
`L<deg>` / `O<deg>`	Set and move to the deadbolt LOCK / OPEN angle
`x`	Both servos limp
`f` / `z`	Flame LEDs for 3 s / buzzer beep
`s`	Status: lid switch, hall sensor, battery voltage, settings
`F`	One full cycle from ARMED: fire, cock, wait for the lid, lock, release, reset

The sketch forgets settings on reboot. Once found, type them into the four lines at its top and upload again.

Power-up order

1. With the battery off, check with the multimeter that +6 V, +5 V and 3V3 each read at least a few hundred ohms to GND. Zero means a short.
2. USB only: send `s`. The lid switch reads `open`; with the gear parked, the hall reads `MAGNET`.
3. Battery on: `z` beeps, `f` lights the flame orange for 3 s, and `s` reports 11–12.6 V.

Calibrate, in this order

Never send `a180` with the lid open and your face over the crown: the gear releases the plunger at about 162°.

1. **Direction and trim.** You found these in Stage 2. Enter `REVERSED` and `TRIM` at the top of the sketch and upload again.
2. **Pick-up.** Gear parked, plunger at its top stop, lid open. Send `a20`. At about 9° the gear catches the rack's first tooth and pulls it down smoothly.
   - If it clicks or the servo stalls at first contact, the tooth tips are meeting. Send `x`, turn the stop grub **4 turns clockwise** (2 mm lower), and try again.
   - If it still clashes, go back to the start and turn it 4 turns anticlockwise instead.
3. **Pull down and hold.** Send `a134`, 2° past the 132° hold notch so the anti-reverse pawl is sure to drop in. The flame drops inside the head, its top about 2–3 mm below the lid's underside. Send `x`. The servo goes limp and the anti-reverse pawl holds the plunger.
4. **Lid and lock.** Close the lid; `s` shows `lid=CLOSED`. Send your lock command, for example `L45`. The bolt slides through the keeper and the lid can no longer be lifted.
5. **Arm.** Send `a180`. Past about 162° the gear lets go and the plunger thumps up against the locked lid. Send `a0`: the servo resets, the gear stays put, and `s` shows `MAGNET` again from the second magnet.
6. **Fire.** Send your open command, for example `O135`. The bolt opens and the flame punches the lid open onto its magnets, standing 44 mm above the crown.
7. **Full cycle.** From ARMED, send `F` and follow the prompts. It should run without a stall or a missed tooth.

Write down for the firmware: `REVERSED`, `TRIM`, `LOCK_DEG`, `OPEN_DEG`, and how many turns you moved the stop grub from nominal.

Final assembly and the first full scare cycle

Join the head to the neck only after the helper sketch's full cycle (`F`) has run cleanly on the bench.

1. **Umbilical into the head.** Thread the five umbilical wires up through the small off-centre hole in the floor of the head's neck cup. Pull them through to the hub without strain.
2. **Head onto the spring.** Lower the head so the neck spring's top coils seat 10 mm deep in the head's cup. Snug its 2 × M3×6 grubs against the top coil. Check the umbilical runs loosely inside the spring and isn't pinched at either cup.
3. **Hub connections.** Solder the umbilical wires to the hub (+6 V, +5 V, VBAT_EXP, both GNDs). Plug in the servos, sensors, LED band and both LCD bundles.
4. **Close the base.** Fit the steel plate with its feet. The base is now 2.3 kg; without the plate the pumpkin tips over on the first shove.
5. **Front mask.** Lay the LCD bundles so nothing rubs the gear, rack or flame. Offer the mask to the seam: the dowels locate it and the magnets pull it shut.
6. **Push test.** Shove the head firmly at eye height. It should rock at about 1.3 swings per second, stop against the tilt ring at about 22°, and settle within a few seconds. The base must not lift.
7. **Armed state.** Run `F` once more with the head on the neck. Leave it ARMED: lid locked, plunger pressing up against the lid, flame hidden.
8. **Charging.** Switch off, plug the 12.6 V charger into the DC jack, and charge to full before the contest. A full pack covers the 3-hour runtime with a wide margin.

The pumpkin is now ready for the show firmware. It reads the four calibration numbers you wrote down.

Troubleshooting

Most faults trace back to three settings: stop-grub height, zero trim and power wiring. Check those first.

Symptom	Likely cause	Fix
Click or stall when the gear first meets the rack	Tooth tips meeting: rack 2 mm out of the clean window	Stop grub 4 turns clockwise (2 mm lower); if still bad, 4 turns anticlockwise from nominal
Lid won't close after pulling down	Plunger slipped back one notch (120° instead of 132°)	Pull down to 134° before going limp; raise `TRIM` by 1°
Gear follows the servo back on reset	Anti-reverse pawl not engaging	Lug must press into the outer teeth; tighten its screw; check the post isn't cracked

Symptom	Likely cause	Fix
Servo turns but the gear doesn't	Drive pawls not reaching the inner teeth	Check the centre screw and washer; gear seated fully on the hub
Gear locks in both directions	A pawl lug too long or too stiff	Sand the lug's sloped face; check the drive disk is flat on the horn
Flame doesn't fire when the bolt opens	Bolt not fully withdrawn, or keeper misaligned	Re-find `OPEN_DEG`; the bolt tip must clear the keeper window
Lid fires open but falls back	Hinge too tight or hold-open magnets reversed	Back the hinge screw out a quarter turn; flip one 6×2 magnet
`hall` never shows `MAGNET`	Magnet facing the wrong way, or sensor on 3.3 V	Flip the gear magnets; A3144 VCC must be +5 V
ESP32 resets when a servo moves	Servo current on thin or shared wires	Servos on the +6 V bus with 18 AWG; 1000 µF at the hub; ESP32 fed only through the 1N5817
MG995 jitters or buzzes at hold	XL4015 current limit too low, or the servo left powered	Current pot fully clockwise; the firmware and `x` go limp once the pawl holds
Flame LEDs dark or wrong colours	Data into the wrong end, or no level shifter	Feed the strip's arrow (DIN) end; 74AHCT125 and 330 Ω in line
LED wire loop catches	Loop too long or too short	Reset it at the top stop: just slack, about 40 mm
Shell halves won't stay closed	A seam magnet glued with the wrong pole out	Find the repelling pair and reverse that one magnet
Base lifts when the head is pushed	Steel plate missing or a smaller plate	230 × 230 × 4 mm plate (1.8 kg) is required